Sebastian K. Müller

Education

2015-2018 Max Planck Institute for Meteorology, Hamburg

Ph.D. Earth System Modeling
Dissertation: *Convectively Generated Gravity Waves and Convective Aggregation in Numerical Models of Tropical Dynamics*
Adviser: Elisa Manzini

2007-2014 Ludwig-Maximilians-University, Munich

M.Sc. Meteorology
Thesis: *Tornadic Supercell Thunderstorms in landfalling Tropical Cyclones*
Adviser: Roger K. Smith

B.Sc. Physics plus Meteorology
Thesis: *Penumbra Functions for Solar Radiation Measurement Instruments*
Adviser: Stefan Wilbert

Work Experience

Expedition Meteorology *03/2023-* PowderGuide.com

Personalized weather forecasts for high-altitude alpinism in the Americas

Mountains:

- Ancohuma, Bolivia [6427m] (27/04/2026)
- Parinacota, Bolivia [6380m] (01/05/2026)
- Chearoco, Bolivia [6127m] (04/05/2026)
- Denali, USA [6190m] (15/05/2026)
- Mount Logan, Canada [5959m] (ongoing)

PostDoc *03/2023-03/2026* University of Trento

Project: next generation Earth System Models

Supervisor : Simona Bordoni

PostDoc *09/2019-08/2022* ICTP, Earth System Physics, Trieste

- Heavy precipitation events in convection-permitting regional climate models
- Tracking Algorithms and Inter-comparison

- Model data postprocessing and model testing

PostDoc — *11/2018-04/2019* — Max Planck Institute for Meteorology, Hamburg

- Convective Aggregation in Sea Surface Temperature Gradients
- Contribution of Simulations to the *Radiative Convective Equilibrium Model Intercomparison Project*
- Tutoring exercises of Prof. Bjorn Stevens' lecture *All about Atmospheric Waves*

Guest Researcher — *03/2017* — University of Tokyo

- Analysis of NICAM-ICSOM1 model output

Student Assistant — *03/2014-03/2015* — Meteorological Institute Munich

Tropical Meteorology working group
- initialization in thermal wind balance for a numerical model
- visualizations of satellite and numerical model data

Student Assistant — *2009-2012* — Meteorological Institute Munich

- remote sensing (LIDAR)
- numerical parameterizations of radiation

Internship — *08-11/2011* — Potsdam Institute for Climate Impact Research

Transdisciplinary Concepts and Methods - Research Domain IV

Internship and B.Sc.-Thesis — *04-10/2010* — German Aerospace Center

Institute of Solar Research at Plataforma Solar de Almeria, Spain

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Key Publications

Segura, H., Müller, S. K., et al. (2025). *nextGEMS: entering the era of kilometer-scale Earth system modeling*. Geoscientific Model Development 18.20 (2025): 7735-7761.

Müller, S. K., Pichelli, E., Coppola, E., et al. (2023). *The climate change response of alpine-mediterranean heavy precipitation events*. Climate Dynamics.

Müller, S. K., Caillaud, C., Chan, S., de Vries, H., et al. (2022). *Evaluation of alpine-mediterranean precipitation events in convection-permitting regional climate models using a set of tracking algorithms*. Climate Dynamics.

Müller, S. K., & Hohenegger, C. (2020). *Self-aggregation of convection in spatially varying sea surface temperatures*. Journal of Advances in Modeling Earth Systems, 12(1), e2019MS001698.

Müller, S. K., Manzini, E.,Giorgetta, M. A., Sato, K., & Nasuno, T. (2018). *Convectively generated gravity waves in high resolution models of tropical dynamics*. Journal of Advances in Modeling Earth Systems, 10, 2564–2588.

Co-Authorships

Coppola, E., Raffaele, F., Giorgi, F., Giuliani, G., Xuejie, G., Ciarlo, J. M., Müller, S. K., . . . & Rechid, D. (2021). Climate hazard indices projections based on CORDEX-CORE, CMIP5 and CMIP6 ensemble. Climate Dynamics, 57(5), 1293-1383.

Wing, A. A., Stauffer, C. L., Becker, T., Reed, K. A., Ahn, M. S., Arnold, N. P., Müller, S. K., . . . & Zhao, M. (2020). *Clouds and convective self-aggregation in a multimodel ensemble of radiative-convective equilibrium simulations*. Journal of advances in modeling earth systems, 12(9), e2020MS002138.

Smith, R. K., Montgomery, M. T., Kilroy, G., Tang, S., and Müller, S. K. (2015): *Tropical low formation during the Australian monsoon: The events of January 2013*, Aust. Meteorol. Oceanogr. J, 65, 318-341.

Douillet, G. A., Taisne, B., Tsang-Hin-Sun, È, Müller, S. K., Kueppers, U., and Dingwell, D. B.: *Syn-eruptive, soft-sediment deformation of deposits from dilute pyroclastic density current: triggers from granular shear, dynamic pore pressure, ballistic impacts and shock waves*, Solid Earth, 6, 553-572, 2015.

Conference Contributions

Presentation EGU General Assembly Conference, 2022: The Climate Response of Heavy Precipitation Events over the Alps and in the Mediterranean.

PICO EGU General Assembly Conference, 2021: The Climate Response of Heavy Precipitation Events over the Alps and in the Mediterranean.

Poster EGU General Assembly Conference, 2018: Sea Surface Temperature Gradients and Convective Aggregation

Poster AGU Fall meeting, 2016: Tropical Convection and Gravity Waves in Spherical Limited Area Models

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Teaching Experience

Tutor *10/2018-04/2019* University of Hamburg

All About Atmospheric Waves
Lecturer: Bjorn Stevens

Tutor *05/2016* International Max Planck Research School for Earth System Modeling, MPI, Hamburg

Introductory Course

Programming Skills

Advanced	python, LaTeX, Unix/Linux
Basic	Fortran, MATLAB, git

Languages

German	Mothertongue
English	Professional
Italian	Fluent
Spanish	Intermediate
French and Latin	Basic

June 16, 2026